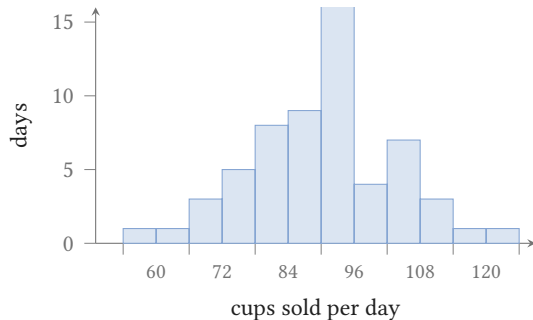


Stage 1

The 1D Gaussian

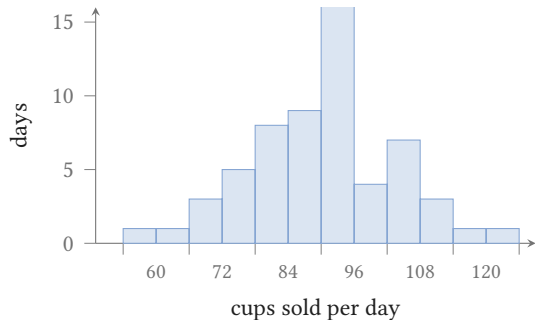


Daily sales, and a question



Real-world observations from 60 days.

Daily sales, and a question



Real-world observations from 60 days.

What is the chance that tomorrow we sell **between 90 and 120 cups**?

Prepare too little and we run out. Prepare too much and it melts.

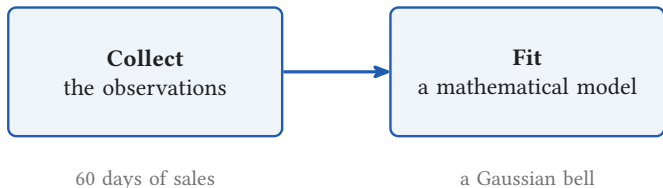
$$P(90 \leq X \leq 120) = ?$$

How data science answers it

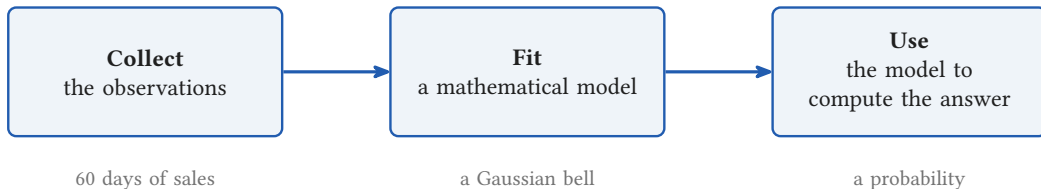
Collect
the observations

60 days of sales

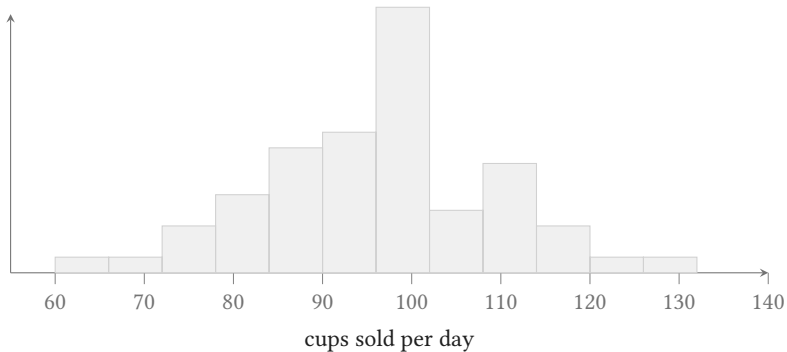
How data science answers it



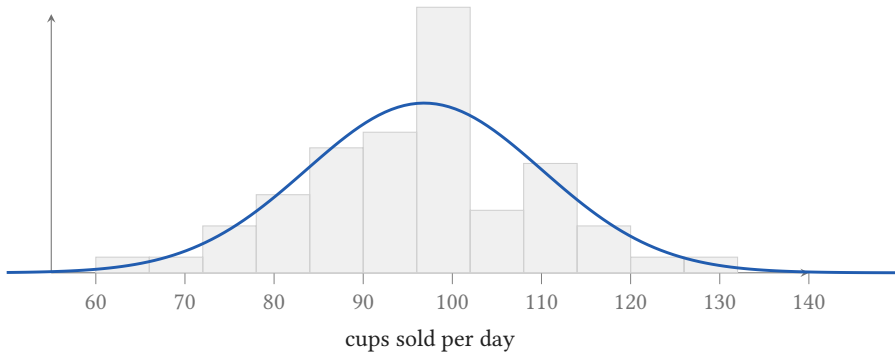
How data science answers it



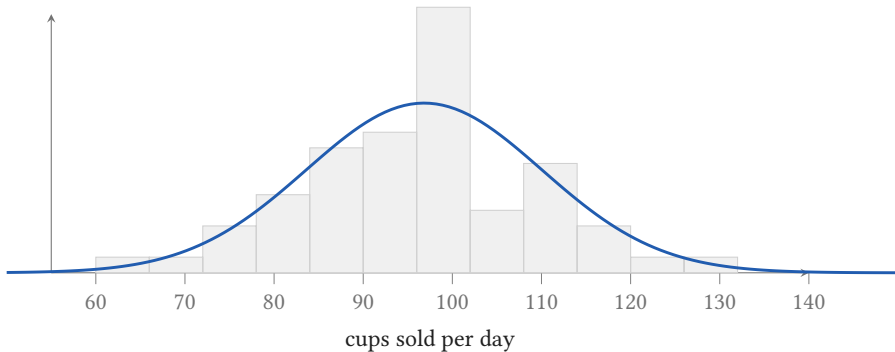
The model is a Gaussian



The model is a Gaussian

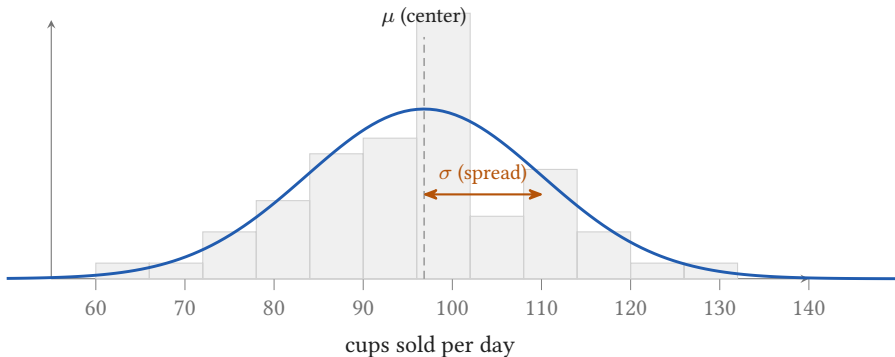


The model is a Gaussian



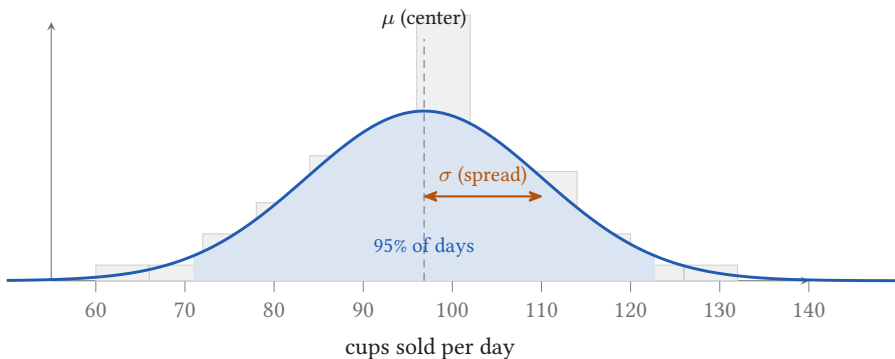
$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The model is a Gaussian



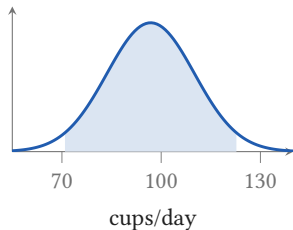
$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

The model is a Gaussian



$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Estimating the center and the spread

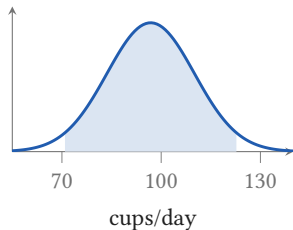


the fitted bell

Average of the data, which settles at the true mean:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \xrightarrow{n \rightarrow \infty} \mathbb{E}[X] =: \mu$$

Estimating the center and the spread



the fitted bell

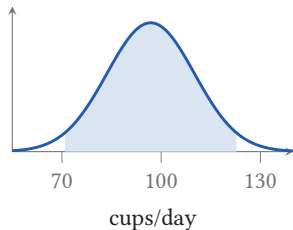
Average of the data, which settles at the true mean:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \xrightarrow{n \rightarrow \infty} \mathbb{E}[X] =: \mu$$

Average squared distance from the center, i.e. the variance:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 \xrightarrow{n \rightarrow \infty} \mathbb{E}[(X - \mu)^2] =: \sigma^2$$

Estimating the center and the spread



the fitted bell

Average of the data, which settles at the true mean:

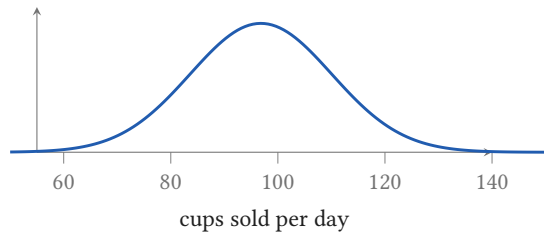
$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \xrightarrow{n \rightarrow \infty} \mathbb{E}[X] =: \mu$$

Average squared distance from the center, i.e. the variance:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 \xrightarrow{n \rightarrow \infty} \mathbb{E}[(X - \mu)^2] =: \sigma^2$$

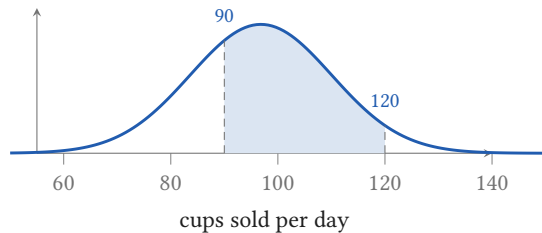
$$\hat{\mu} = 96.8 \quad \hat{\sigma}^2 = 174.8$$

Answering the question



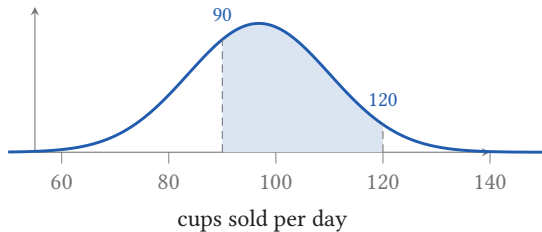
$$P(90 \leq X \leq 120) = \int_{90}^{120} g(x) dx$$

Answering the question



$$P(90 \leq X \leq 120) = \int_{90}^{120} g(x) dx$$

Answering the question



$$P(90 \leq X \leq 120) = \int_{90}^{120} g(x) dx$$

$$P = 0.66$$

about a 2-in-3 chance

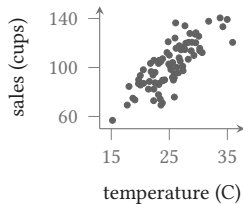
observations $\rightarrow$ a Gaussian model $\rightarrow$ a probability we
can plan against

Stage 2

The 2D Gaussian

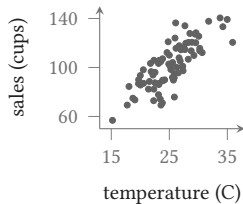
2

Three worlds, and a question

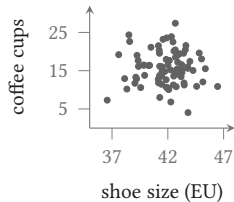


hotter $\rightarrow$ more sales

Three worlds, and a question

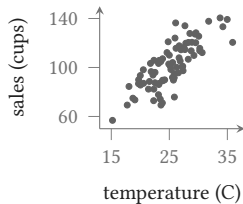


hotter → more sales

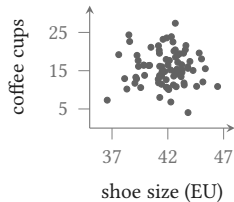


no relationship

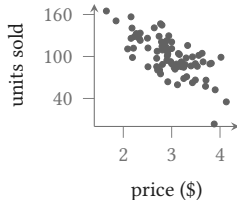
Three worlds, and a question



hotter $\rightarrow$ more sales

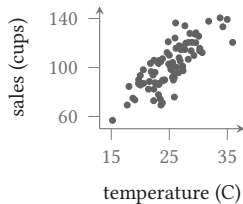


no relationship

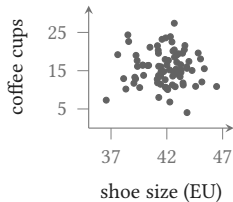


pricier $\rightarrow$ fewer sold

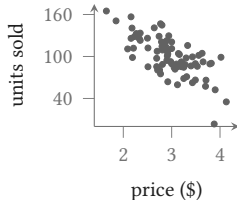
Three worlds, and a question



hotter $\rightarrow$ more sales



no relationship



pricier $\rightarrow$ fewer sold

On the first world, what is the chance a day is **warm and busy at once**?

temperature between 25 and 30 C, sales between 100 and 140 cups

$$P\left(T \in [25, 30] \text{ and } S \in [100, 140] \right) = ?$$

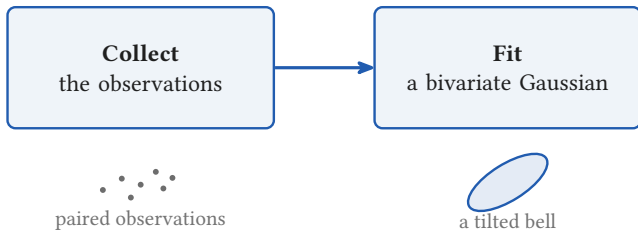
The same method, now in 2D

Collect
the observations



paired observations

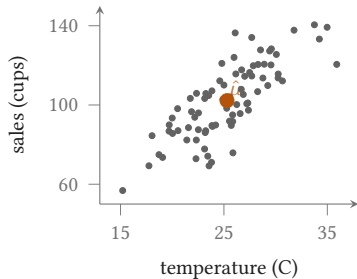
The same method, now in 2D



The same method, now in 2D

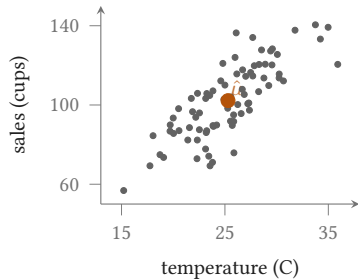


Estimating the pieces: center and per-axis spread



Idea: run Stage 1 once on each axis.

Estimating the pieces: center and per-axis spread



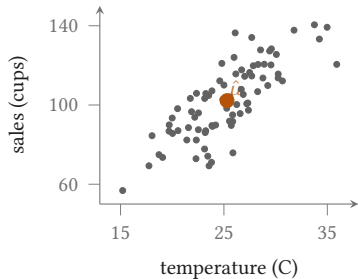
the mean, marked

Idea: run Stage 1 once on each axis.

The mean vector (average of each coordinate):

$$\hat{\mu} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = \begin{pmatrix} 25.3 \\ 102.2 \end{pmatrix}$$

Estimating the pieces: center and per-axis spread



the mean, marked

Idea: run Stage 1 once on each axis.

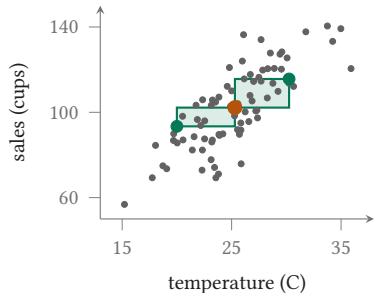
The mean vector (average of each coordinate):

$$\hat{\mu} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = \begin{pmatrix} 25.3 \\ 102.2 \end{pmatrix}$$

The spread on each axis (variance):

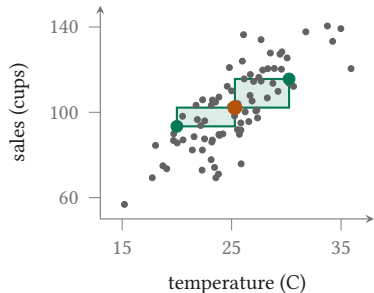
$$\hat{\sigma}_1^2 = 16.0 \text{ (temperature)} \quad \hat{\sigma}_2^2 = 348.2 \text{ (sales)}$$

Estimating the pieces: the correlation



two sample rectangles; signed area = a product

Estimating the pieces: the correlation

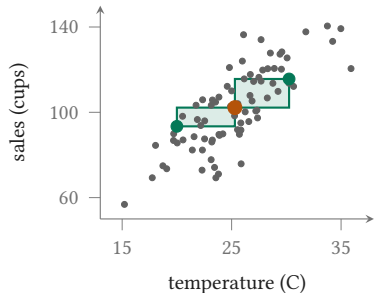


Average of those signed areas:

$$\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^n (x_{1i} - \hat{\mu}_1)(x_{2i} - \hat{\mu}_2)$$
$$\xrightarrow{n \rightarrow \infty} \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)]$$

two sample rectangles; signed area = a product

Estimating the pieces: the correlation



two sample rectangles; signed area = a product

Average of those signed areas:

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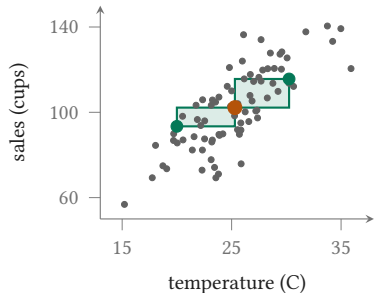
Its sign is the headline, across the three worlds:

positive $\hat{\sigma}_{12} = +58.8$, corr $+0.79$

neutral $\hat{\sigma}_{12} \approx 0$, corr 0.00

negative $\hat{\sigma}_{12} = -9.7$, corr -0.70

Estimating the pieces: the correlation



two sample rectangles; signed area = a product

Average of those signed areas:

$$\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^n (x_{1i} - \hat{\mu}_1)(x_{2i} - \hat{\mu}_2)$$

$$\xrightarrow{n \rightarrow \infty} \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)]$$

Its sign is the headline, across the three worlds:

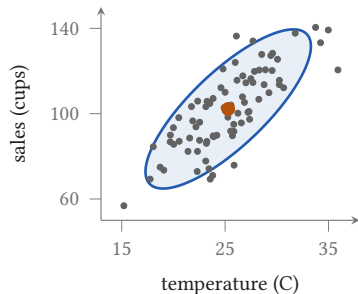
positive $\hat{\sigma}_{12} = +58.8$, corr $+0.79$

neutral $\hat{\sigma}_{12} \approx 0$, corr 0.00

negative $\hat{\sigma}_{12} = -9.7$, corr -0.70

The scale-free reading is the correlation, always in $[-1, 1]$.

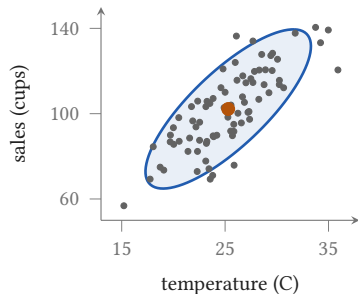
The bivariate Gaussian, and its covariance matrix



with the point $\mathbf{x} = (x_1, x_2)^\top$ and center $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$:

the fitted bell, tilted by + correlation

The bivariate Gaussian, and its covariance matrix

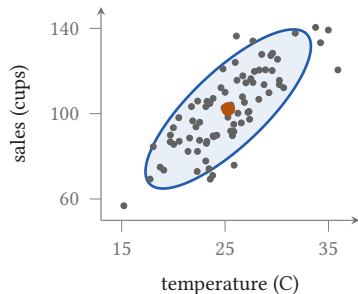


the fitted bell, tilted by + correlation

with the point $\mathbf{x} = (x_1, x_2)^\top$ and center $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$:

$$g(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

The bivariate Gaussian, and its covariance matrix



the fitted bell, tilted by + correlation

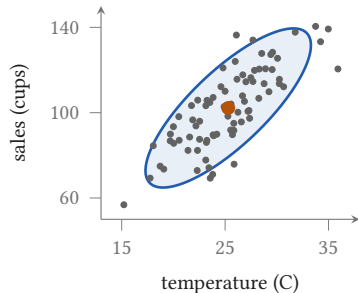
with the point $\mathbf{x} = (x_1, x_2)^\top$ and center $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$:

$$g(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Σ is the covariance matrix: spreads on the diagonal, covariance off it.

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$$

The bivariate Gaussian, and its covariance matrix



the fitted bell, tilted by + correlation

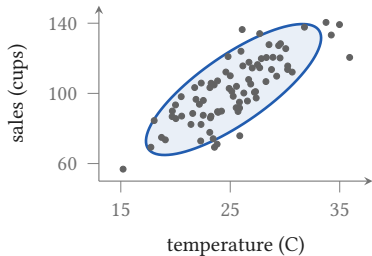
with the point $\mathbf{x} = (x_1, x_2)^\top$ and center $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$:

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Σ is the covariance matrix: spreads on the diagonal, covariance off it.

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} = \begin{pmatrix} 16.0 & 58.8 \\ 58.8 & 348.2 \end{pmatrix}$$

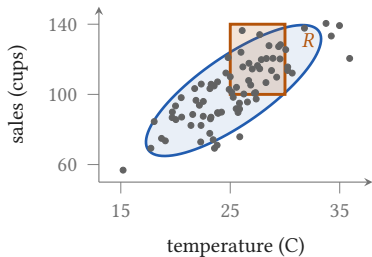
The answer: a double integral over the region



Chance the day lands inside the rectangle R :

the region $R = [25, 30] \times [100, 140]$

The answer: a double integral over the region

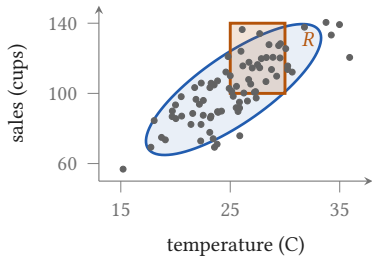


Chance the day lands inside the rectangle R :

$$P((X_1, X_2) \in R) = \iint_R g(x_1, x_2) dx_1 dx_2$$

the region $R = [25, 30] \times [100, 140]$

The answer: a double integral over the region



the region $R = [25, 30] \times [100, 140]$

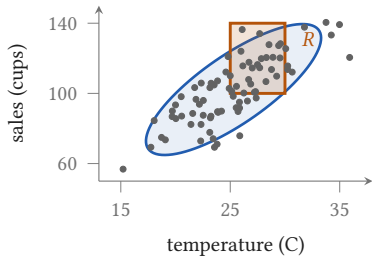
Chance the day lands inside the rectangle R :

$$P((X_1, X_2) \in R) = \iint_R g(x_1, x_2) dx_1 dx_2$$

$$P = 0.31$$

about a 1-in-3 chance

The answer: a double integral over the region



the region $R = [25, 30] \times [100, 140]$

Chance the day lands inside the rectangle R :

$$P((X_1, X_2) \in R) = \iint_R g(x_1, x_2) dx_1 dx_2$$

$$P = 0.31$$

about a 1-in-3 chance

two questions at once $\rightarrow$ a bivariate Gaussian (μ, Σ from data) $\rightarrow$ a probability over a region

Stage 3

The multivariate Gaussian

3

Three flavors over 40 days, and a question

day	vanilla	chocolate	strawberry
1	55	41	35
2	64	44	36
3	41	29	30
4	67	36	39
5	59	41	35
6	73	53	37
⋮	⋮	⋮	⋮

Each row is one day; each number is cups sold. Hot days lift all three flavors together.

Each number is cups sold that day. 40 days in total.

Three flavors over 40 days, and a question

day	vanilla	chocolate	strawberry
1	55	41	35
2	64	44	36
3	41	29	30
4	67	36	39
5	59	41	35
6	73	53	37
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Each number is cups sold that day. 40 days in total.

Each row is one day; each number is cups sold. Hot days lift all three flavors together.

We plan a target range for each flavor. Call a day **balanced** when all three land inside their target at once, so nothing runs out and nothing is left over:

vanilla 40 to 60, chocolate 30 to 50, strawberry 20 to 40 cups

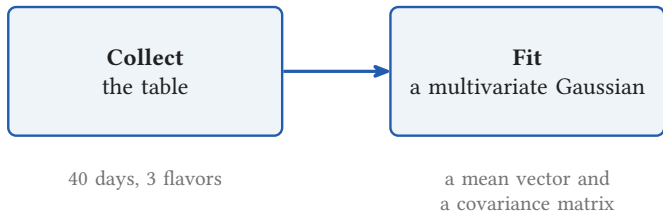
$$P(\text{a balanced day}) = ?$$

The same method, in three dimensions

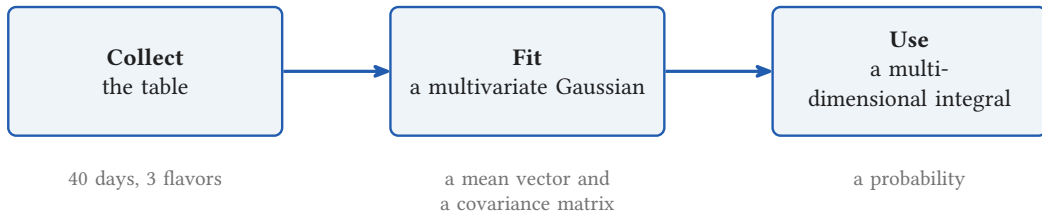
Collect
the table

40 days, 3 flavors

The same method, in three dimensions



The same method, in three dimensions



A mean vector and a covariance matrix

The mean vector, one average per flavor (per column of the table):

$$\hat{\mu} = (53.4, 41.5, 31.4) \quad (\text{vanilla, chocolate, strawberry), in cups}$$

A mean vector and a covariance matrix

The mean vector, one average per flavor (per column of the table):

$$\hat{\mu} = (53.4, 41.5, 31.4) \quad (\text{vanilla, chocolate, strawberry}), \text{ in cups}$$

The covariance matrix $\hat{\Sigma}$ (spreads on the diagonal):

	vanilla	chocolate	strawberry
vanilla	91.6	66.7	51.8
chocolate	66.7	108.0	49.6
strawberry	51.8	49.6	51.1

all nine entries positive: the flavors move together.

A mean vector and a covariance matrix

The mean vector, one average per flavor (per column of the table):

$$\hat{\boldsymbol{\mu}} = (53.4, 41.5, 31.4) \quad (\text{vanilla, chocolate, strawberry), in cups}$$

The covariance matrix $\hat{\Sigma}$ (spreads on the diagonal):

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vanilla	91.6	66.7	51.8
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strawberry	51.8	49.6	51.1

all nine entries positive: the flavors move together.

The general Gaussian density, for a point $\mathbf{x}$ in d dimensions:

$$g(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \sqrt{|\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

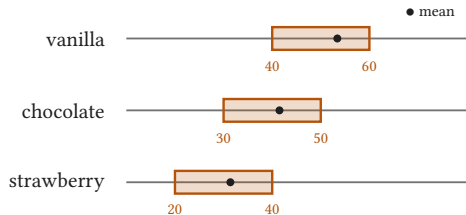
Here $d = 3$. (Stage 1 was $d = 1$, Stage 2 was $d = 2$; only the size of $\boldsymbol{\mu}$ and Σ changes.)

The answer: a multi-dimensional integral



Chance every flavor lands in its target range at once (a balanced day):

The answer: a multi-dimensional integral



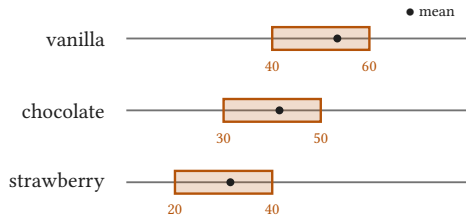
each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

$$P(\mathbf{X} \in R) = \iiint_R g(x_1, x_2, x_3) dx_1 dx_2 dx_3$$

the balanced region $R = [40, 60] \times [30, 50] \times [20, 40]$ cups

The answer: a multi-dimensional integral



each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

$$P(\mathbf{X} \in R) = \iiint_R g(x_1, x_2, x_3) dx_1 dx_2 dx_3$$

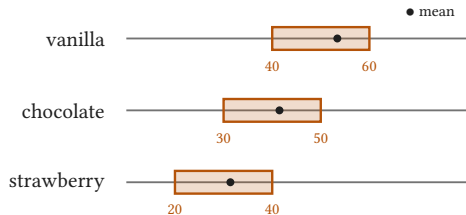
the balanced region $R = [40, 60] \times [30, 50] \times [20, 40]$ cups

$$P = 0.48$$

about half our days are balanced; the rest miss on a flavor

The multivariate Gaussian

The answer: a multi-dimensional integral



each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

$$P(\mathbf{X} \in R) = \iiint_R g(x_1, x_2, x_3) dx_1 dx_2 dx_3$$

the balanced region $R = [40, 60] \times [30, 50] \times [20, 40]$ cups

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about half our days are balanced; the rest miss on a flavor

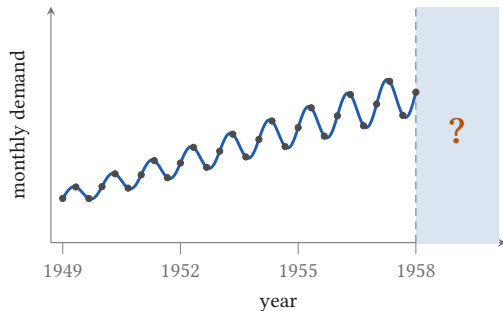
many quantities at once $\rightarrow$ a multivariate Gaussian (μ vector, Σ matrix) $\rightarrow$ a probability over a region

Stage 4

The Gaussian process



Demand through 1958, and a question

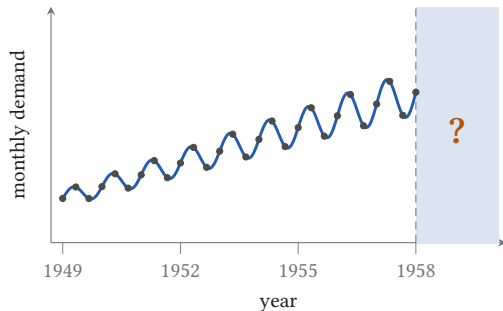


We track a monthly **demand** series through 1958.

Real airline-passenger numbers stand in for a shop's revenue series.

Monthly demand through 1958; the next two years are blank.

Demand through 1958, and a question



Monthly demand through 1958; the next two years are blank.

We track a monthly **demand** series through 1958.

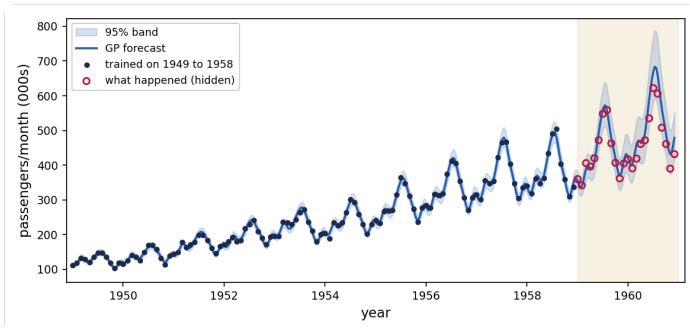
Real airline-passenger numbers stand in for a shop's revenue series.

What will demand be over the **next two years**?

And how **sure** can we be?

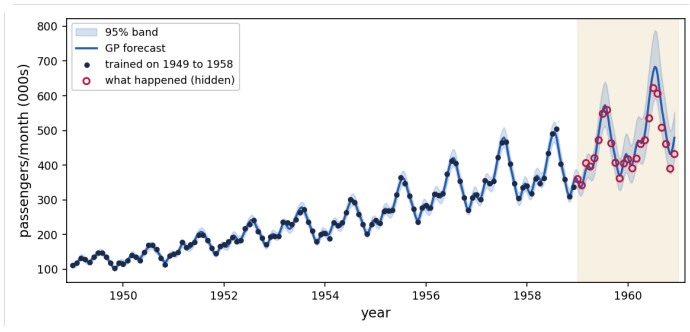
We want an honest range, not just a single number.

Fit a Gaussian process: the forecast



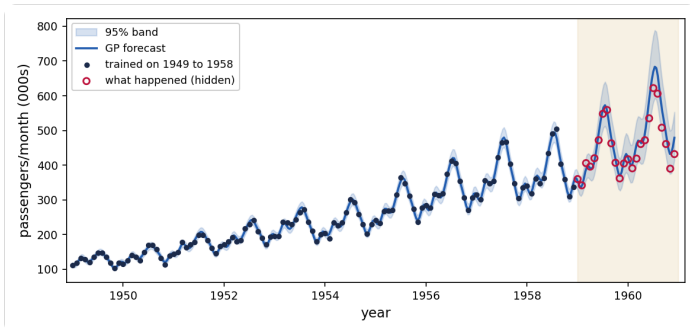
Trained only on the **navy dots**: monthly demand, 1949 to 1958.

Fit a Gaussian process: the forecast



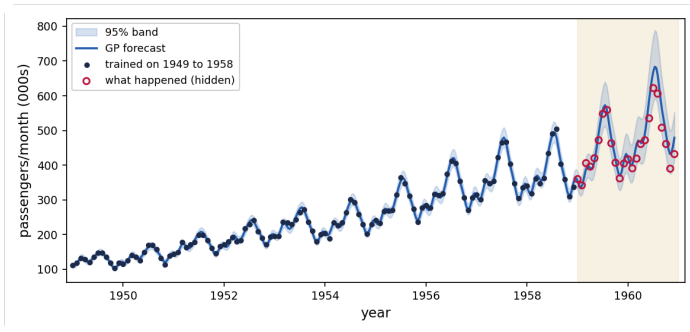
The **blue line and band** are the two-year forecast that follows.

Fit a Gaussian process: the forecast



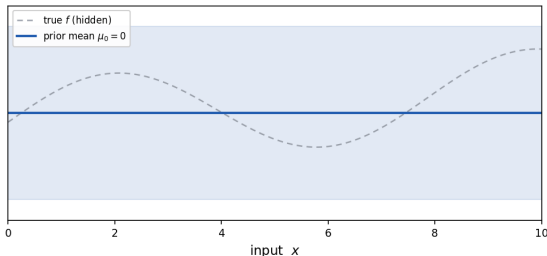
The **red circles** are the hidden truth, every one **inside the band** ($\approx 4.6\%$ error).

Fit a Gaussian process: the forecast



A forecast that also says how sure it is.

So what is a Gaussian process?



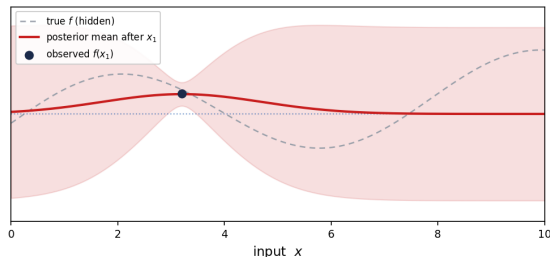
A Gaussian process is just **two** things:

1. A mean function $m(x) = \mathbb{E}[f(x)]$

the best guess at every input (flat before data).

Before any data: the mean is a **flat guess** μ_0 and the band is **wide**. We know almost nothing.

So what is a Gaussian process?



A Gaussian process is just **two** things:

1. A mean function $m(x) = \mathbb{E}[f(x)]$

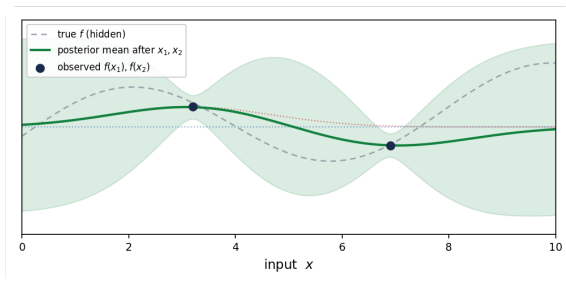
the best guess at every input (flat before data).

2. A kernel $k(x, x')$

how similar two inputs are; nearby inputs $\rightarrow$ correlated,
so the mean bends and the band pinches at the data.

Pick x_1 , observe $f(x_1)$: the mean **bends** to it, and the band **pinches** at the point.

So what is a Gaussian process?



Pick x_2 , observe $f(x_2)$: the mean **threads both**, and the band tightens. More data, less doubt.

A Gaussian process is just **two** things:

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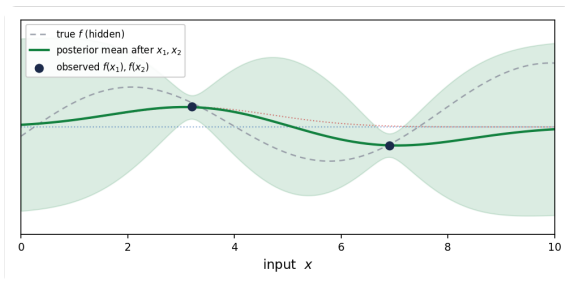
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The **band** is the leftover uncertainty: small at the data, wide away.

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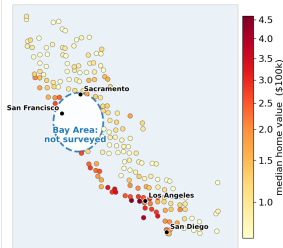
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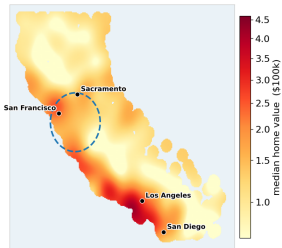
We infer m and k by Bayesian inference, which we do not cover here; the library does it.

The same idea, over a map

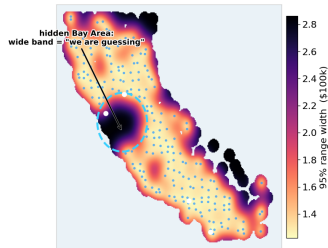
1. What we surveyed
189 scattered areas (avg. home value)



2. The model fills in the whole state
best estimate of home value everywhere



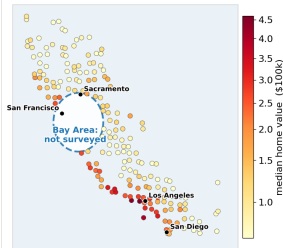
3. How sure is it? width of the 95% range
tight where we surveyed, wide where we did not



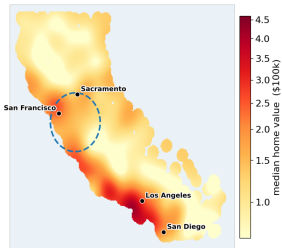
Now the input is a **place**; the kernel compares two locations.

The same idea, over a map

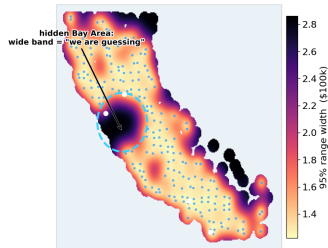
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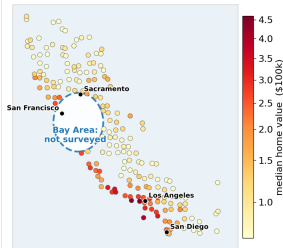
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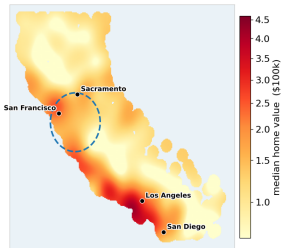
From a scattered survey, one Gaussian process **fills in the whole state**.

The same idea, over a map

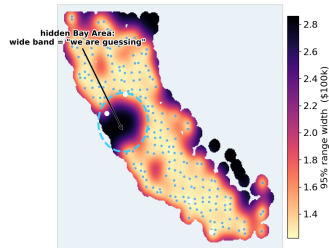
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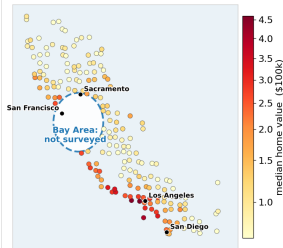
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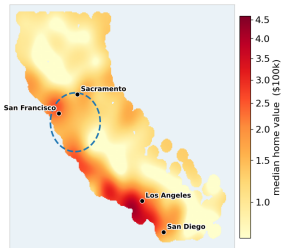
Panel 3 flags the guessing: $\approx 1.8x$ wider over the Bay Area, still **88% covered**.

The same idea, over a map

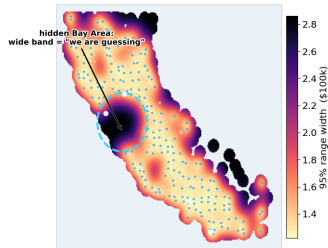
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tight where we surveyed, wide where we did not



The same Gaussian process, a different input space.

One idea, many input spaces

A time
a demand forecast

input = a time

One idea, many input spaces

A time
a demand forecast

input = a time

A place
a filled-in map

input = a place

One idea, many input spaces

A time
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input = a time

A place
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A set of objects
a cocktail, a molecule

the finale, still to come

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In code it is a few lines: choose a kernel, fit, then predict with a band.

One idea, many input spaces

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input = a time

A place
a filled-in map

input = a place

A set of objects
a cocktail, a molecule

the finale, still to come

In code it is a few lines: choose a kernel, fit, then predict with a band.

a mean function + a kernel $\rightarrow$ predict anywhere, with
honest error bars

Stage 5

A Gaussian process over anything

A Gaussian process over anything

One idea, any object

A Gaussian process needs only two things: a **mean** and a **kernel** (a similarity).

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For an object, the kernel asks one thing: **how much do two objects share?**

One idea, any object

A Gaussian process needs only two things: a **mean** and a **kernel** (a similarity).



For an object, the kernel asks one thing: **how much do two objects share?**

three acts, all of them objects:



pizzas: topping sets



emojis: tag sets



molecules: fingerprints

Pizza: the object is a set of toppings



a pizza = its set of toppings.

20 possible toppings, so $2^{20} =$

1,048,576

possible pies. We bake only a few.

Pizza: the object is a set of toppings



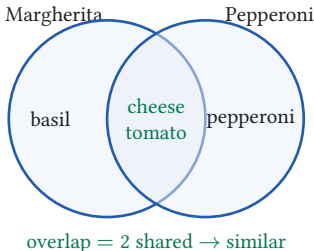
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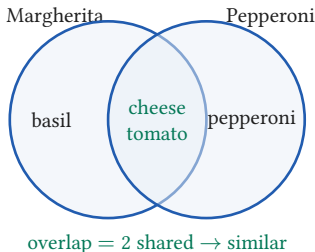
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A dessert pie {nutella, banana, sugar} shares 0 with Margherita → far.

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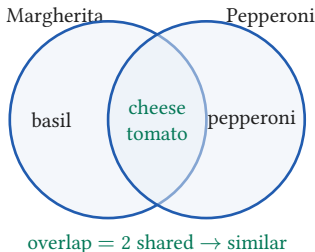
20 possible toppings, so $2^{20} =$

1,048,576

possible pies. We bake only a few.

Rate a few; the Gaussian process predicts *all* of them, with error bars: tight near what we tested, wide for strange combos.

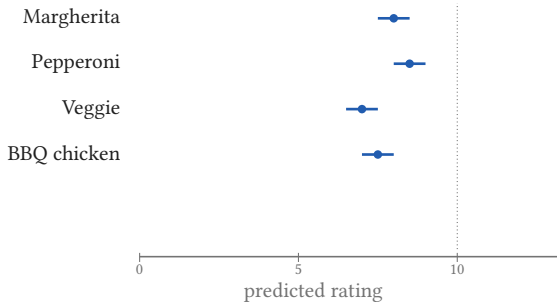
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A Gaussian process over anything

Pizza: predicting a pie we never baked

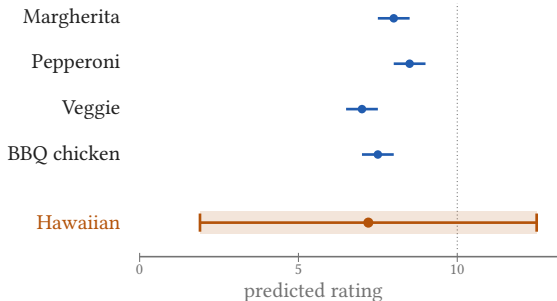


We have **never baked Hawaiian** (ham + pineapple).

each bar is a predicted rating with its error bar.

A Gaussian process over anything

Pizza: predicting a pie we never baked



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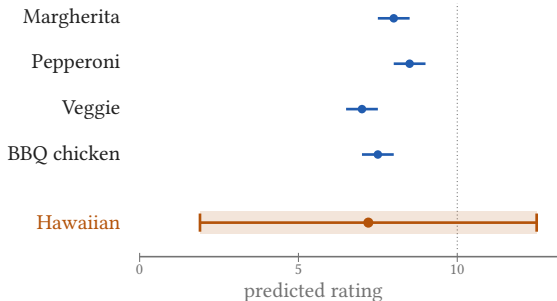
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The GP still predicts a rating, **7.2** ...

...but with a very wide band, [1.9, 12.5].

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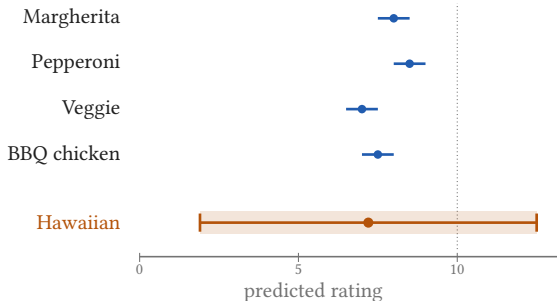
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Pineapple is unlike anything we have tested, so we honestly do not know.

Pizza: predicting a pie we never baked



each bar is a predicted rating with its error bar.

The pineapple war, settled with an error bar.

We have **never baked Hawaiian** (ham + pineapple).

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...but with a very wide band, [1.9, 12.5].

Pineapple is unlike anything we have tested, so we honestly do not know.

So test this pie next (a hidden gem or a disaster) = smart experimentation.

A Gaussian process over anything

Emoji: tags, and a question



+18

hot, lit, trending, hype



+15

launch, growth,
trending, hype



+12

new, shiny,
special, trending



+6

cute, please,
soft, emotional



+2

dead, done, extreme, tired

Each emoji ships with official Unicode keyword **tags** the moment it is released; f = its open-rate lift in a subject line.

A Gaussian process over anything

Emoji: tags, and a question



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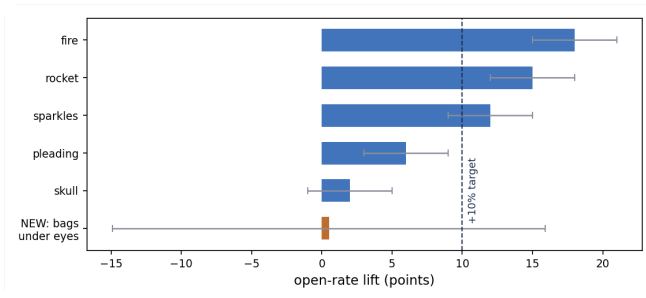


Unicode 16.0 just shipped “**face with bags under eyes**”, with **zero posts** anywhere.

Before we send a single email: $P(\text{open-rate lift} \geq 10\%)$?

A Gaussian process over anything

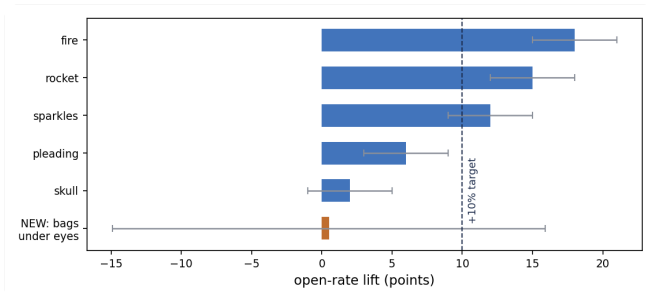
Emoji: the answer, before any post



From its **tags alone**, the Gaussian process scores the never-posted emoji.

A Gaussian process over anything

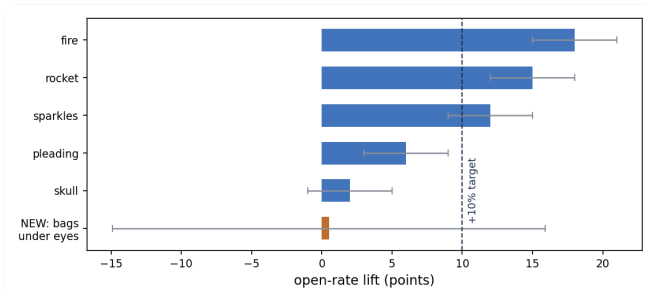
Emoji: the answer, before any post



Bags-under-eyes overlaps only **skull** (weak) and nothing hype $\rightarrow$ predicted ≈ 0 , wide band $\rightarrow$
 $P(\geq 10\%) = 11\% \rightarrow$ **skip it.**

A Gaussian process over anything

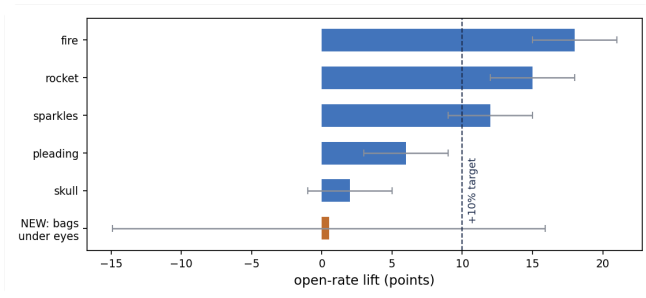
Emoji: the answer, before any post



Flip side: a hype-tagged new emoji {trending, hype, viral} sits by **fire** and **rocket** → predicted **+13** → $P = 73\%$ → **headline it**.

A Gaussian process over anything

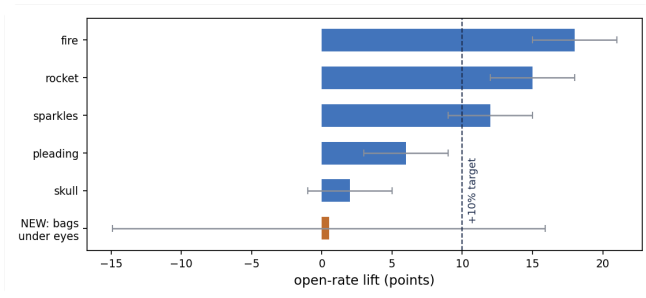
Emoji: the answer, before any post



Same machine, opposite verdict, before either was ever posted.

A Gaussian process over anything

Emoji: the answer, before any post



Then a Bayesian-optimization loop picks which handful of 3,000+ emojis to A/B-test next.

Molecule: the object, and the stakes



a molecule = its **structural fingerprint** (the set of substructures it contains).

~3,000,000

could be made; almost none ever smelled.

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Kernel = **Tanimoto** overlap on fingerprints. The same overlap idea as pizza toppings and emoji tags, and a proven proper kernel.
 f = sandalwood-ness, on a 0 to 10 scale.

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Real money: natural sandalwood is CITES-listed and over-harvested, so a cheap synthetic that *out-smells* it is worth millions. This exact loop runs at Givaudan, Firmenich, IFF.

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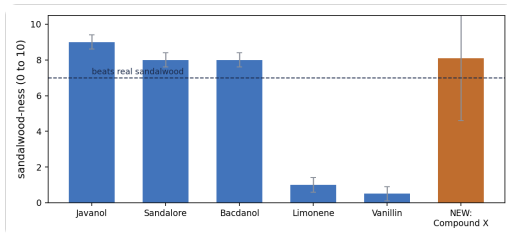
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We can make and smell only ~20 of 3,000,000.
Which 20?

A Gaussian process over anything

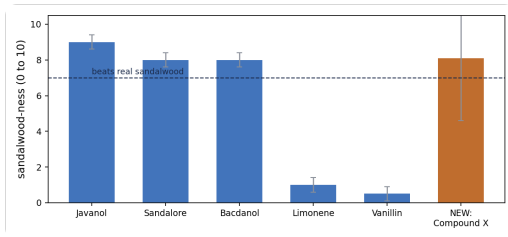
Molecule: the answer, and the close



Compound X: a molecule **no one has ever made**, a cousin of three known sandalwoods.

A Gaussian process over anything

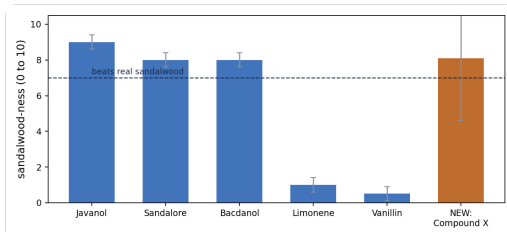
Molecule: the answer, and the close



Predicted **8.1/10** $\rightarrow P(\text{beats real sandalwood}) = 73\% \rightarrow$ **synthesize it.**

A Gaussian process over anything

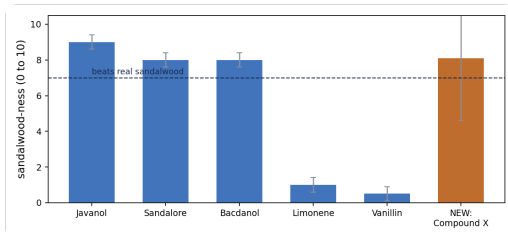
Molecule: the answer, and the close



Bayesian optimization narrows 3,000,000 candidates to a couple of syntheses.

A Gaussian process over anything

Molecule: the answer, and the close



From a pizza topping to a molecule: one kernel.

a mean + a kernel $\rightarrow$ honest predictions over any objects

numbers, places, pizzas, emojis, molecules · from data to solutions, over anything.